Python Dictionary

Create a Dictionary

```
# creating a dictionary
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
  "England": "London"
}

# printing the dictionary
print(country_capitals)
```

```
{'Germany': 'Berlin', 'Canada': 'Ottawa', 'England': 'London'}
```

Access Dictionary Items

```
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
  "England": "London"
}

# access the value of keys
print(country_capitals["Germany"])    # Output: Berlin
print(country_capitals["England"])    # Output: London
```

```
Berlin
London
```

Add Items to a Dictionary

```
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
}
```

```
# add an item with "Italy" as key and "Rome" as its value
country_capitals["Italy"] = "Rome"

print(country_capitals)
```

```
{'Germany': 'Berlin', 'Canada': 'Ottawa', 'Italy': 'Rome'}
```

Remove Dictionary Items

```
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
}

# delete item having "Germany" key
del country_capitals["Germany"]

print(country_capitals)
```

```
{'Canada': 'Ottawa'}
```

remove all items

```
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
}

# clear the dictionary
country_capitals.clear()

print(country_capitals)
```

```
{}
```

Change Dictionary Items

```
country_capitals = {
  "Germany": "Berlin",
  "Italy": "Naples",
  "England": "London"
}

# change the value of "Italy" key to "Rome"
country_capitals["Italy"] = "Rome"

print(country_capitals)
```

```
{'Germany': 'Berlin', 'Italy': 'Rome', 'England': 'London'}
```

Iterate Through a Dictionary

```
country_capitals = {
  "United States": "Washington D.C.",
  "Italy": "Rome"
}

# print dictionary keys one by one
for country in country_capitals:
    print(country)

print()

# print dictionary values one by one
for country in country_capitals:
    capital = country_capitals[country]
    print(capital)
```

```
United States
Italy

Washington D.C.
Rome
```

Find Dictionary Length

```
country_capitals = {"England": "London", "Italy": "Rome"}
```

```
# get dictionary's length
print(len(country_capitals))   # Output: 2

numbers = {10: "ten", 20: "twenty", 30: "thirty"}

# get dictionary's length
print(len(numbers))            # Output: 3

countries = {}

# get dictionary's length
print(len(countries))          # Output: 0
```

```
2
3
0
```

Dictionary Membership Test

```
file_types = {
    ".txt": "Text File",
    ".pdf": "PDF Document",
    ".jpg": "JPEG Image",
}

# use of in and not in operators
print(".pdf" in file_types)       # Output: True
print(".mp3" in file_types)       # Output: False
print(".mp3" not in file_types)   # Output: True
```

```
True
False
True
```